
Introduction to Healthcare in the Context of Ageing Societies: A Structured Essay

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Healthcare is changing rapidly: technology is evolving, and new treatments are being implemented at a breakneck pace on a global scale. However, healthcare still remains limited in many aspects. One critical aspect is health and wellbeing for the elderly population. While technology enabling effective care for vulnerable populations has increased, many needs remain unmet. Among them are access to healthcare, effective nutrition and food technology, availability of tools for medication management and rehabilitation, support for mental health, as well as community-driven care. This booklet aims to explore these topics briefly and provide a foundation to solve issues at large.

NOTE: This text was written as a part of the SCORE Research Camp x Hackathon 2025/26. It is intended as only a brief overview of the topic for hackathon participants. Please perform your own research to support ideas being pitched. Furthermore, please do not limit innovations to the topics discussed within this document. Additionally, as a matter of principle, no large language models were used in the authorship process of this document.

1 Ageing Societies

In the 21st century, both developed and developing countries are rapidly ageing. The percentage of citizens aged over 60 (defined as “elderly citizens” *Ageing Society VS. Aged Society in Global and Thai Contexts* n.d.) is rising in almost all developed countries worldwide. This demographic trend poses a challenge for a wide multitude of sectors, including finance, entertainment, and healthcare. Thailand is no exception to the ageing society trend, as it is rapidly transitioning into a super-aged society, with 28% of its population expected to be aged 60 or older by 2030. This shift is transforming every aspect of Thai society—from healthcare and financial systems to economic productivity and family structures.

This transition does not occur in a vacuum: these aspects interact with each other and often result in great additive effects. One commonly cited example in economics is the indirect effect of an ageing population on the labour supply, which is best explained by the requirements imposed resulting in a restructuring of roles within the family unit (where working-aged adults are moved away from the labour force

Table 1: Age-Based Categories for Societies Aging Society VS. Aged Society in Global and Thai Contexts n.d.

Society Category	Definition
Ageing society	Age 60+ > 10% of pop OR Age 65+ > 7% of pop
Aged society	Age 65+ > 14% of pop
Super-aged society	Age 65+ > 20% of pop

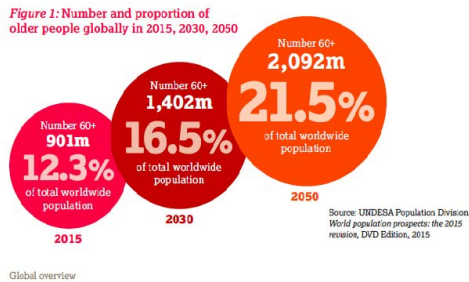


Figure 1: Composition of the world's population by age Aging Society VS. Aged Society in Global and Thai Contexts n.d.

towards caregiving). While not directly relevant to the present context of this hackathon, such effects do not limit their influence to merely the economic sector and labour force. Healthcare as a sector is also heavily affected as well. The direct and indirect effects of healthcare from these factors is left as an exercise to the reader.

2 Thailand's Healthcare System

Thailand's healthcare system is overall considered one of the best globally in preparedness for pandemic and epidemic diseases, with a rank of 5 out of 195 countries according to *Thailand - GHS Index* (2026). This composite ranking considers a holistic perspective on the healthcare system as a whole, including elements such as the accessibility of healthcare, the availability of prehospital care, disease response capability and detection capacity among other metrics. However, these metrics paint a different story than the picture that most Thais (and HCWs) see of the healthcare system on a daily basis, with overcrowding, long wait times and, in distant districts, a distinct lack of HCWs, as readiness for pandemics does not indicate how well the system functions in everyday practice.



Figure 2: The typical patient load in public hospitals *Talking with Pictures: Going to the Hospital - Pantip* 2026

2.1 Inequity of Healthcare

The availability of healthcare within countries is not distributed evenly. The highest proportion of healthcare workers (HCWs) tend to be found in the most developed areas. This is especially true for Thailand, where the “primate city” of Bangkok serves as a hub for a large number of world-class healthcare facilities, while remote provinces often receive care that may be considered as below the standard set by big cities. In fact, over four million people in Thailand alone are without ready access to healthcare services Tilakul, 2026. Factors driving this inequity inem, though it has some limitations, ranging from the risk of decreasing equity of care, to the limited tec healthcare include the lack of homogeneity in development, the selection pool for training of HCWs, uneven distribution of funding and human factors among others. While initiatives such as those by the NHSO exist to allow patients access to hospital, they are often found to have limited scope in practice, resulting in ineffective solutions to the problem of healthcare access. This is shown best in figure 3, which shows how doctors are distributed unevenly to the patient population, and that workload differs greatly across healthcare districts. One potential solution involves increasing funding and availability of healthcare centers in distant provinces, however, the feasibility of such solutions often depend on individual community context. Consequently, a comprehensive solution remains notably absent.

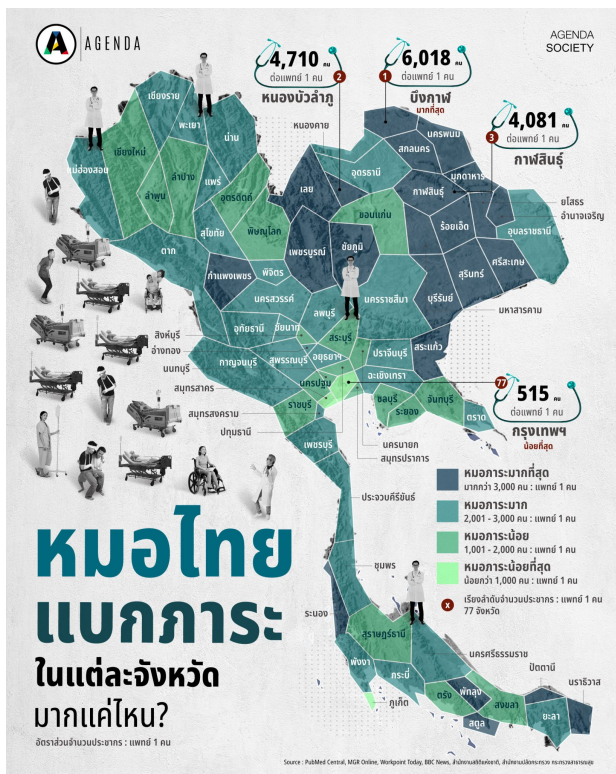


Figure 3: Distribution of healthcare availability across healthcare districts in Thailand *How Heavy Is the Burden Borne by Thai Doctors in Each Province? 2022*

2.2 Funding Public Projects

As the population ages, the proportion of the working class continually decreases, resulting in a decreased amount of revenue collected in taxes. While taxes are always controversial, such decreases as a result of a shrinking worker demographic undeniably results in less funds spent in the development of infrastructure and other essential public utilities. These factors could increase the prevalence of preventable diseases such as waterborne agents, mosquito populations and other sanitation based maladies, leading an already overloaded healthcare system at the brink of collapse. Furthermore, social welfare systems, including in Thailand, are at great risk as governments rush to cut spending and adapt to the challenge presented. More information on this topic may be found here: *Thailand's Ageing Society Threatens Fiscal Collapse and Drags down Economy* (2025), as well as in socioeconomics and the “dependency ratio.”

With these considerations, many policy projects focus on how to enable governments to function in this state. However, the details of such projects and their context is beyond the context of this article. However,

meaningful and cost-conscious solutions to this slowly developing problem are sorely lacking, especially in the healthcare sector and healthcare services delivery.

3 Social Context

Healthcare is more than just treating symptoms and prescribing medicines for illness. Effective healthcare demands the involvement of HCWs in the community, both at the individual and community scale. Consequently, community relationships serve as a key method for the prevention of disease and to increase the availability of healthcare within communities, especially those in remote areas. Thailand is unique in this context, with a large network of community HCWs known as “village health volunteers (VHVs)” who serve to protect the health of their communities, monitor vulnerable residents and effectively report community health status to the relevant authorities to promote informed policy decisions that reflect community status at both the community and country scale by narrowing the gap between policymakers and residents of the local area. (One way to define the decentralization of healthcare is the term “community based care”)

Community based care is an effective system, though it has some limitations, including the risk of decreasing equity of care, to the limited technology available for VHVs. Solutions to these problems are few and far between, but could hold the potential to be highly effective and beneficial for the community at large.

3.1 Expectations of Healthcare

Thailand healthcare system is accessible to many, which is often touted as a great success of the design. However, this is accompanied by large patient loads, primarily as a result of ease of access. According to Pannarunothai (2014), the number of outpatient visits per hundred thousand population hovers around 30,000 in every year, indicating significant resource utilization and benefit to the population. However, ease of access also gives rise to outpatient visits that may not be deemed strictly necessary. While Thailand is aiming to cut these visits by public initiatives such as incorporating pharmacy prescriptions into the *gold card* program, allowing for basic prescriptions to be dispensed without the need to visit the hospital. Con-

sidering the current situation, more and timely solutions are required to maintain the quality of essential hospital functions, which the government is processing. Some solutions that have not yet been implemented in full scale projects include telehealth solutions, which have origins from the *Coronavirus* pandemic. However, private sector solutions may also be considered for implementation in the present context.

3.2 The Mental Health Stigma

As a society, Thailand is developed in some aspects, and below international standards on others. One major factor where Thailand requires improvement is the mental health stigma: the majority of Thais harbour negative opinions towards receiving mental health treatment for fear of a social stigma. While social change is slow, it is direly needed to ensure that the population is able and *especially willing* to receive treatment, especially as the country ages and the ropes of the loneliness epidemic tighten day-by-day.

3.3 Healthy Habits

One challenge to providing effective preventative healthcare for the general population is the lack of action regarding healthy eating habits. The typical Thai meal does not cover all nutritional bases, and often exceeds healthy levels for many macronutrients, including calories, carbohydrates, fats, salt and sugar. The habit of choosing food that is highly flavorful and relatively in excess compared to the nutritional requirements may result in long-term consequences such as the increased risk for NCDs, including renal, cardiovascular and diabetes related complications. With this context, or even without, it can be stated that behavioral change is difficult. Therefore, interventions must effectively understand factors influencing behaviour and the *why* behind the way the brain works. Those interested in this topic should look at Loucks et al. (2019), Halpern et al. (2015), and Davidson and Scholz (2020), which are major studies. There is also information available on the “Science of Behaviour Change” page on the NIH common fund (<https://commonfund.nih.gov/behaviorchange>).

4 Challenges for the Elderly

4.1 Non-communicable Diseases

Non-communicable disease refers to chronic disease that is contracted on an individual basis. Common NCDs include diabetes mellitus (type 2), hypertension, dyslipidemia, which result in increased risk for comorbidities including cardiovascular disease, coronary or peripheral artery disease and various ischemias. While NCDs have a genetic component, many cases occur as a result of a behavioral etiology. Consequently, behavioral change is an effective treatment in a majority of NCDs. Present interventions receive limited success, as they have implementation challenges that are often not well circumvented. Please see section 3.3 for details on effective intervention design.

4.2 Life Quality & Interventions

The quality of life of a population is determined by a wide range of factors, including social environment, region, social equity and personal factors among others. Though over 30 years has passed, the Dahlgren and Whitehead model remains one of the standard ways of understanding the influences of health even before a visit to the medical practitioner. Many solutions for wellbeing do not start in the hospital, but instead in the community, where interventions are often more effective than treatment of already developed conditions. The prevention of disease, especially non-communicable diseases remains more cost effective than treatment. However, many limitations remain that result in social interventions being overlooked. Solutions in this field are direly needed, and may serve to actively enable the prevention of diseases in the subclinical phase, resulting in healthier communities and longer healthspans for the population.

Any effective interventions on life quality for the population, especially in the elderly population, must take into account the feasibility of these interventions in actual practice. Many a lesson has been learned by well-meaning NGOs and even governmental aid organizations that intervention effectiveness is not determined by the quality of each product or commercial, but the feasibility of adoption to daily life is the factor which indicates if interventions will prove successful in the long run. One example of this is the well-meaning diabetes diet and exercise regimens, which

often face failure as a consequence of how patients are not able to adhere to them, and often end up reverting to their original, disease causing, diets. Therefore, one could speculate that more realistic interventions based on individual ability could drive better compliance and greater success Sami et al., 2017. Put simply, behavioral friction kills effective interventions and technology. For further details on policy interventions, please see Weaver (n.d.).

5 Emerging Technologies

Overall, artificial intelligence (AI) is emerging as a powerful tool for change in healthcare systems and technology. From AI text transformers to advanced data processing and analytics, AI holds the potential to improve access to healthcare and increase the efficiency of Thailand's medical infrastructure. However, several barriers still hinder the widespread adoption of AI in healthcare. These include technological complexity, lack of cultural sensitivity, and a failure to address the specific needs of the Thai elderly population and their families. Solutions to enable the effective and ethical utilization of artificial intelligence, especially generative artificial intelligence are needed to combat this, and enable the utilization of these technologies for good.

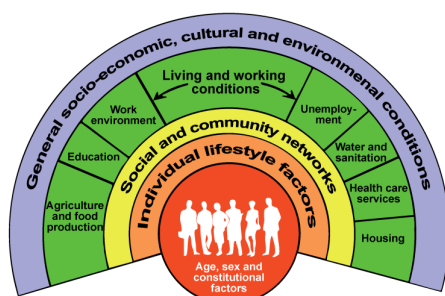


Figure 4: Model of the social determinants of health, originally by Dahlgren and Whitehead, 1991, from Mariette-JB, 2026

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